

MATH 20700 Honors Analysis I

Homework 6

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Problem 4.27

Proof. (a) A contraction is a mapping $f : M \rightarrow M$ such that for some constant $k < 1$ and $\forall x, y \in M$, we have $d(fx, fy) \leq k \cdot d(x, y)$.

A weak contraction is a mapping $f : M \rightarrow M$ such that $\forall x, y \in M$, if $x \neq y$ then $d(fx, fy) < d(x, y)$.

We observe that the contraction constant k is uniform for all $x, y \in M$, whereas for a weak contraction, the ratio $\frac{d(fx, fy)}{d(x, y)} < 1$ but could vary depending on x, y and is not uniformly bounded by a constant upper bound (i.e., in principle, for a weak contraction $\frac{d(fx, fy)}{d(x, y)}$ could approach $0.999 \dots$)

Hence, a weak contraction is not a contraction.

Note that by Mean Value Theorem, weak contraction is the same as $\forall x \in M, |f'(x)| < 1$. And similarly by Mean Value Theorem, contraction is the same as $\forall x \in M, |f'(x)| \leq k < 1$.

For a concrete counterexample, consider $f : \mathbb{R} \rightarrow \mathbb{R}$ defined as $f(x) = \ln(e^x + 1)$. Then, $f'(x) = \frac{e^x}{e^x + 1}$, and we have that $\forall x \in \mathbb{R}, f'(x) < 1$ but $\sup_{x \in \mathbb{R}} |f'(x)| = \lim_{x \rightarrow \infty} f'(x) = 1$.

Then, there does not exist a uniform constant k so that $\forall x \in \mathbb{R}, |f'(x)| \leq k < 1$, so it is a weak contraction but not a contraction.

(b) Define the points

$$x_0 := 0, \quad x_n := 1 - 2^{-n} \quad (n \geq 1),$$

and the intervals $I_n := [x_{n-1}, x_n]$ for $n \geq 1$. For each $n \geq 1$ set the slope

$$s_n := 1 - 2^{-n} \in (0, 1).$$

Define $f : [0, 1] \rightarrow [0, 1]$ by $f(0) := 0$ and, for $x \in I_n$,

$$f(x) := f(x_{n-1}) + s_n(x - x_{n-1}).$$

This makes f continuous and strictly increasing on $[0, 1]$.

Moreover

$$f(1) := \lim_{x \rightarrow 1^-} f(x) = \sum_{n=1}^{\infty} s_n |I_n| = \sum_{n=1}^{\infty} (1 - 2^{-n}) 2^{-n} = \sum_{n=1}^{\infty} 2^{-n} - \sum_{n=1}^{\infty} 4^{-n} = 1 - \frac{1}{3} = \frac{2}{3} \in [0, 1],$$

so indeed $f : [0, 1] \rightarrow [0, 1]$.

Let us check that f is a weak contraction. If $x, y \in I_n$, then $|f(x) - f(y)| = s_n|x - y| < |x - y|$. If $x < y$ lie in several consecutive intervals, choose points

$$x = t_0 < t_1 < \cdots < t_m = y$$

so that each $[t_{j-1}, t_j] \subset I_{n_j}$. Then, by the triangle inequality and the previous one-interval estimate,

$$|f(y) - f(x)| \leq \sum_{j=1}^m |f(t_j) - f(t_{j-1})| = \sum_{j=1}^m s_{n_j}(t_j - t_{j-1}) < \sum_{j=1}^m (t_j - t_{j-1}) = y - x.$$

Thus $|f(x) - f(y)| < |x - y|$ for all $x \neq y$.

Let us show that f is not a contraction.

Suppose for sake of contradiction that there exists $L < 1$ with $|f(x) - f(y)| \leq L|x - y|$ for all x, y . Choose n so large that $s_n = 1 - 2^{-n} > L$ and take distinct $x, y \in I_n$. Then

$$|f(x) - f(y)| = s_n|x - y| > L|x - y|,$$

a contradiction. Hence no global $L < 1$ works and f is not a contraction.

- (c) First let us show existence. Define $g(x) := d(f(x), x)$, then g is continuous on a compact domain, hence it attains its minimum and maximum. Let us call the point where the minimum occurs x_0 . So, $g(x_0) \leq g(x)$ for all $x \in M$.

We want to find a point $y \in M$ such that $g(y) = 0$. Suppose the minimum value of the function is strictly greater than zero, i.e., $g(x_0) > 0$. Then, $d(f(x_0), x_0) > 0$, hence $g(f(x_0)) = d(f(f(x_0)), f(x_0)) < d(f(x_0), x_0) = g(x_0)$ contradicting the assumption that x_0 is the point where the minimum of g occurs. Hence $g(x_0) = 0$ (i.e., the minimum of g is 0), hence x_0 is a fixed point of f .

Uniqueness is trivial. Suppose $x \neq y$ and $f(x) = x$ and $f(y) = y$ (i.e., we have 2 distinct fixed points). Then, $d(fx, fy) = d(x, y) < d(x, y)$ is a contradiction.

QED

Problem 4.30

Proof. Let S^1 be the unit circle in $\mathbb{R}^2$ and consider the continuous map $f : S^1 \rightarrow S^1$ defined by $f(v) = -v$.

A fixed point v must satisfy the equation $f(v) = v$, which implies $-v = v$. This equation's only solution in $\mathbb{R}^2$ is the zero vector, $v = 0$. However, the zero vector is not an element of the unit circle S^1 , as all points $v \in S^1$ must satisfy $\|v\| = 1$. Since the only possible fixed point is not in the space, the map has no fixed point.

QED

Problem 4.41

Proof. (a) Define the function $f : [0, 1] \times [0, 1] \rightarrow \mathbb{R}$ as:

$$f(x, y) = \begin{cases} \frac{xy}{x^2+y^2} & \text{if } (x, y) \neq (0, 0) \\ 0 & \text{if } (x, y) = (0, 0). \end{cases}$$

Then, as $(x, y) \rightarrow (0, 0)$ along the axes (i.e., as $\lim_{y \rightarrow 0}(0, y)$ or as $\lim_{x \rightarrow 0}(x, 0)$), the numerator is always 0, hence $f(x, y) = 0$.

However, as $(x, y) \rightarrow (0, 0)$ along the line $y = x$, we have $\lim_{(x,y) \rightarrow (0,0)} f(x, y) = \lim_{(x,y) \rightarrow (0,0)} \frac{xy}{x^2+y^2} = \lim_{(x,y) \rightarrow (0,0)} \frac{x^2}{2x^2} = \lim_{(x,y) \rightarrow (0,0)} \frac{1}{2} = \frac{1}{2}$.

Further, we observe that if we fix $x = c$ with $c \in [0, 1]$, then $f(x, y) = \frac{cy}{c^2+y^2}$ which is continuous, and vice versa if we fix $y = c$.

(b) Let (x_0, y_0) be an arbitrary point in the domain, then we want to show that f is continuous at (x_0, y_0) .

Fix $\epsilon > 0$, then we want to show that $\exists \delta > 0$ such that if $d((x, y), (x_0, y_0)) < \delta$ then $|f(x, y) - f(x_0, y_0)| < \epsilon$.

We can write $|f(x, y) - f(x_0, y_0)| \leq |f(x, y) - f(x_0, y)| + |f(x_0, y) - f(x_0, y_0)|$.

By equicontinuity of $\mathcal{E}$, $\exists \delta_1 > 0$ such that if $d((x, y), (x_0, y)) < \delta_1$ then $|f(x, y) - f(x_0, y)| < \frac{\epsilon}{2}$.

Since for each fixed x , the function $y \rightarrow f(x, y)$ is a continuous function of y , we know that $\exists \delta_2 > 0$ such that if $d((x_0, y), (x_0, y_0)) < \delta_2$ then $|f(x_0, y) - f(x_0, y_0)| < \frac{\epsilon}{2}$.

Then, choosing $\delta = \min\{\delta_1, \delta_2\}$ completes the proof.

QED

Problem Prelim 1

Proof. Fix $x \in \mathbb{R}$. Suppose for sake of contradiction that f is not continuous at x . Then there exists a sequence (y_n) with $y_n \rightarrow x$ but $f(y_n) \not\rightarrow f(x)$. Hence there exists $\epsilon > 0$ and a subsequence (y_{n_k}) such that

$$|f(y_{n_k}) - f(x)| \geq \epsilon \quad \text{for all } k.$$

For each fixed k , consider the constant sequence $x_n \equiv y_{n_k}$. By the hypothesis, $f_n(y_{n_k}) \rightarrow f(y_{n_k})$ as $n \rightarrow \infty$. Thus we can choose m_k inductively so that $m_k > m_{k-1}$ (with m_1 arbitrary) and

$$|f_{m_k}(y_{n_k}) - f(y_{n_k})| < \frac{\epsilon}{2}.$$

Define a new sequence (z_n) by

$$z_n = \begin{cases} y_{n_k}, & n = m_k, \\ x, & \text{otherwise.} \end{cases}$$

Since $z_n = x$ except along the indices m_k , and $y_{n_k} \rightarrow x$, it follows that $z_n \rightarrow x$.

Applying the hypothesis to (z_n) , we get $f_n(z_n) \rightarrow f(x)$. However, along the subsequence $n = m_k$,

$$|f_{m_k}(z_{m_k}) - f(x)| = |f_{m_k}(y_{n_k}) - f(x)| \geq |f(y_{n_k}) - f(x)| - |f_{m_k}(y_{n_k}) - f(y_{n_k})| \geq \epsilon - \frac{\epsilon}{2} = \frac{\epsilon}{2},$$

so $f_{m_k}(z_{m_k})$ stays a fixed distance from $f(x)$ and cannot converge to $f(x)$. This contradicts $f_n(z_n) \rightarrow f(x)$. Therefore f is continuous at x . Since x was arbitrary, f is continuous on $\mathbb{R}$.

QED

Problem Prelim 2

Proof. Let us prove that the sequence is equicontinuous.

We have $f_{n+1}(x) \leq f_n(x)$ for all x . Since f_n converges monotonically down to the constant (continuous) function 0, by Dini's theorem (proved in Homework 5), we have that the convergence is uniform. Hence, fix $\epsilon > 0$, then $\exists N_\epsilon \in \mathbb{N}$ such that $\forall n \geq N_\epsilon, \forall x \in [a, b], |f_n(x) - 0| < \epsilon$.

This handles the 'tail' part with uniform $\delta_{tail} = b - a$ that works for all f_n 's.

There are finitely many f_n 's with $n < N_\epsilon$, and each of them is continuous on a compact domain so each of them is uniformly continuous. Choose f_k with $k < N_\epsilon$, then it has a corresponding δ_k for uniform continuity of f_k .

Picking $\delta = \min_{k < N_\epsilon} \{\delta_k\}$ completes the proof (clearly $\delta \leq \delta_{tail}$ hence this δ works for equicontinuity of the f_n 's).

QED

Problem Prelim 12

Proof. By FTC, $f(t) = f(0) + \int_0^t f'(x) dx = \int_0^t f'(x) dx$.

Taking absolute values and applying Cauchy-Schwarz, we get $|f(t)| = \left| \int_0^t f'(x) dx \right| = \left| \int_0^t f'(x) \cdot 1 dx \right| \leq \left(\int_0^t (f'(x))^2 dx \right)^{1/2} \cdot \left(\int_0^t (1)^2 dx \right)^{1/2} = \left(\int_0^t (f'(x))^2 dx \right)^{1/2} \cdot \sqrt{t} \leq \left(\int_0^1 (f'(x))^2 dx \right)^{1/2}$.

Above for the last inequality, we observe that $\int_0^t (f'(x))^2 dx \leq \int_0^1 (f'(x))^2 dx$ and $\sqrt{t} \leq 1$.

Thus, squaring both sides, we get $|f(t)|^2 \leq \int_0^1 (f'(x))^2 dx$. Since this holds for all $t \in [0, 1]$, we can substitute the $\|f\|^2$ in place of $|f(t)|^2$ to get the desired inequality.

QED

Problem Prelim 13

Proof. The derivative is bounded, so each f_n is Lipschitz continuous: $|f_n(y) - f_n(x)| \leq 2|y - x|$.

Then, $|g(y) - g(x)| \leq |g(y) - f_n(y)| + |f_n(y) - f_n(x)| + |f_n(x) - g(x)|$. Since this holds for all n , let us take the limit $n \rightarrow \infty$, then we get $|g(y) - g(x)| \leq |f_n(y) - f_n(x)| \leq 2|y - x|$ (the other terms vanish due to the pointwise convergence $f_n \rightarrow g$).

Thus, fix $\epsilon > 0$ as the ‘desired precision’ for continuity of g , then picking $\delta < \frac{\epsilon}{2}$ completes the proof. (In fact, we have shown that g is not just continuous, but Lipschitz continuous).

QED

Problem 5.1

Proof. (a) $\implies$ (b)

We want to show that for any $\epsilon > 0$, we can find a $\delta > 0$ such that if $\|q - p\|_V < \delta$, then $\|T(q) - T(p)\|_W < \epsilon$.

Since T is linear, we can write $T(q) - T(p) = T(q - p)$.

By continuity of T at the origin, we know that $\exists \delta > 0$ such that if $\|x - 0\|_V = \|x\|_V < \delta$, then $\|T(x) - T(0)\|_W = \|T(x)\|_W < \epsilon$.

Thus, setting $x = q - p$, we observe that if $\|q - p\|_V < \delta$, then $\|T(q) - T(p)\|_W = \|T(q - p)\|_W < \epsilon$.

(b) $\implies$ (c)

$p \in V$ and T is continuous at p , so clearly T is continuous at at least point of V (namely, p).

(c) $\implies$ (a)

If T is continuous at at least point of V , $\exists m \in V$ such that T is continuous at m .

Thus, fix $\epsilon > 0$, then $\exists \delta > 0$ such that if $\|q - m\|_V < \delta$, then $\|T(q) - T(m)\|_W = \|T(q - m)\|_W < \epsilon$.

Let $x = q - m$, then we have shown that if $\|x\|_V = \|x - 0\|_V < \delta$, then $\|T(x)\|_W = \|T(x) - T(0)\|_W < \epsilon$. This proves that T is continuous at the origin.

QED

Problem 5.2

Proof. The operator norm for $T \in \mathcal{L}$ is defined as $\|T\|_{op} = \sup_{v \in V} \frac{\|T(v)\|_W}{\|v\|_V} = \sup_{v \in V, \|v\|_V=1} \|T(v)\|_W$.

We want to show that this meets the 3 properties of a norm: positive definiteness, absolute homogeneity, and the triangle inequality.

1. Positive Definiteness

$\|T\|_{op}$ is the supremum of non-negative values $\|T(v)\|_W$, so $\|T\|_{op} \geq 0$.

If $T = \mathbf{0}$, then $\|T\|_{op} = \sup \|\mathbf{0}(v)\|_W = 0$. Conversely, if $\|T\|_{op} = 0$, then $T(v) = \mathbf{0}_W$ for all unit vectors v . By linearity, $T(x) = \|x\|_V T(\frac{x}{\|x\|_V}) = \mathbf{0}_W$ for all $x \in V$, so $T = \mathbf{0}$.

2. Absolute Homogeneity

We must show $\|\alpha T\|_{op} = |\alpha| \|T\|_{op}$. By definition, $\|\alpha T\|_{op} = \sup_{\|v\|_V=1} \|(\alpha T)(v)\|_W = \sup_{\|v\|_V=1} \|\alpha(T(v))\|_W$.

Using the homogeneity of the norm $\|\cdot\|_W$, this becomes $\sup_{\|v\|_V=1} (|\alpha| \|T(v)\|_W)$. Since $|\alpha|$ is a non-negative constant, it factors out of the supremum, yielding $|\alpha| \sup_{\|v\|_V=1} \|T(v)\|_W = |\alpha| \|T\|_{op}$.

3. Triangle Inequality

We must show $\|T + S\|_{op} \leq \|T\|_{op} + \|S\|_{op}$. By definition, $\|T + S\|_{op} = \sup_{\|v\|_V=1} \|(T + S)(v)\|_W = \sup_{\|v\|_V=1} \|T(v) + S(v)\|_W$.

Applying the triangle inequality for the norm $\|\cdot\|_W$ gives $\|T + S\|_{op} \leq \sup_{\|v\|_V=1} (\|T(v)\|_W + \|S(v)\|_W)$.

Since $\|T(v)\|_W \leq \|T\|_{op}$ and $\|S(v)\|_W \leq \|S\|_{op}$ for all such v , the term inside the supremum is bounded by $\|T\|_{op} + \|S\|_{op}$. Therefore, the supremum itself is also bounded, so $\|T + S\|_{op} \leq \|T\|_{op} + \|S\|_{op}$.

QED

Problem 5.8

Proof. (a) **Linearity**

$$T(f) + T(g) = \int_0^x f(t) dt + \int_0^x g(t) dt = \int_0^x f(t) + g(t) dt = \int_0^x (f + g)(t) dt = T(f + g).$$

$$T(cf) = \int_0^x (cf)(t) dt = \int_0^x cf(t) dt = c \int_0^x f(t) dt = cT(f).$$

Continuity It suffices to show that the operator norm $\|T\|_{op}$ is finite.

We observe that

$$\begin{aligned} \|T\|_{op} &= \sup_{f \in C_0, f \neq 0} \frac{\|T(f)\|}{\|f\|} \\ &= \frac{\sup_x \left| \int_0^x f(t) dt \right|}{\|f\|} \\ &\leq \frac{\sup_x \int_0^x |f(t)| dt}{\|f\|} \\ &\leq \frac{\sup_x \int_0^x \|f\| dt}{\|f\|} \\ &= \frac{\|f\| \cdot \sup_x \int_0^x dt}{\|f\|} \\ &= \sup_x \int_0^x dt \\ &= \sup_x x. \end{aligned}$$

This implies that the space C^0 isn't just "functions" but "functions on a specific domain." Given the integral starts at 0, a standard assumption is that C^0 means $C^0([0, L])$, the space of continuous functions on some interval $[0, L]$ (where $L > 0$).

Then, $\sup_x x = L$, hence $\|T\|_{op} \leq L$. This shows that the operator norm $\|T\|_{op}$ is bounded by a finite number, hence T is continuous.

Norm

We know that $\|T\|_{op} \leq L$. Let us show that $\|T\|_{op} = L$. It suffices to find a single $g \in C_0([0, L])$ such that $\frac{\|T(g)\|}{\|g\|} = L$.

Let $g(t) = 1, \forall t \in [0, L]$.

Then, the norm is $\|g\| = \sup_{t \in [0, L]} |1| = 1$.

The output function is $(Tf)(x) = \int_0^x 1 dt = x$.

The norm of the output function is $\|Tf\| = \sup_{x \in [0, L]} |(Tf)(x)| = \sup_{x \in [0, L]} x = L$.

The ratio $\frac{\|Tf\|}{\|f\|} = L$.

Hence, we conclude that $\|T\|_{op} \leq L$.

(b)

$$\begin{aligned} T(f_n) &= \int_0^x f_n(t) dt \\ &= \int_0^x \cos(nt) dt \\ &= \left[\frac{1}{n} \sin(nt) \right]_0^x \\ &= \frac{1}{n} \sin(nx). \end{aligned}$$

- (c) We claim that there are no Cauchy sequences or subsequences in K , hence there are no convergent sequences or subsequences. This is because we can pick arbitrary $n \in \mathbb{N}$, then $\|f_n - f_{2n}\| = \sup_t |\cos(nt) - \cos(2nt)|$. At $t = \pi/n$, the value is $|\cos(\pi) - \cos(2\pi)| = |-1 - 1| = 2$. Thus, K has no limit points, hence it is closed.

K is bounded because clearly $\|f_n\| = \sup_{t \in [0, L]} |\cos(nt)| = 1$ for all n . Since $\|f_n\| \leq 1$ for every function in the set K , the set is bounded.

K is not compact. This is because clearly, the sequence $\cos(nt)$ does not have a convergent subsequence by the above argument.

- (d) $T(K) = \{\frac{1}{n} \sin(nt) \mid n \in \mathbb{N}\}$ is not closed because the uniform limit function is the constant 0 function. (As $n \rightarrow \infty$, $\frac{1}{n} \sin(nt) \rightarrow 0$ uniformly). Hence it is not compact.

We claim that the closure of $T(K)$ is compact. Invoking the Arzela Ascoli theorem, we observe that it suffices to show that the sequence of functions (g_n) with $g_n = \frac{1}{n} \sin(nt)$ is uniformly equicontinuous.

Clearly, each g_n is uniformly continuous, because for any x, y , $|g_n(x) - g_n(y)| < \frac{2}{n}$.

Hence for all indices $n > N := \frac{2}{\epsilon}$, the δ for uniform continuity is ∞ . For indices $n \leq N$, there are finitely many corresponding δ_n so take $\delta^* = \min\{\delta_n\}_{n < N}$. Then, this δ^* works for uniform equicontinuity of the sequence (g_n) .

In fact, we can construct an explicit δ^* . For any $g_n = \frac{1}{n} \sin(nt)$, we observe that the derivative $g'_n = \cos(nt)$ is bounded, i.e., $\forall t, |g'_n(t)| \leq 1$. Hence, applying Mean Value Theorem, we observe that for any x, y , $|g_n(x) - g_n(y)| = |\sin(x) - \sin(y)| \leq 1 \cdot |x - y|$, hence $\delta^* = \epsilon$ shows uniform equicontinuity of (g_n) .

QED